Chapter 7

More maps and spaces

7.0.1 Annihilators

The annihilator provides a precise duality between subspaces of V and subspaces of V^*.

Definition 7.0.1: Annihilator

For a subspace $U \subseteq V$, the **annihilator** of U is

$$\mathrm{Ann}(U) = U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}.$$

Theorem 7.0.1 Dimension of the annihilator

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(\mathrm{Ann}(U)) = \dim(V) - \dim(U).$$

Proof: The restriction map $\rho : V^* \to U^*$ defined by $\rho(\ell) = \ell|_U$ is a linear map with $\ker(\rho) = \mathrm{Ann}(U)$. We claim ρ is surjective: given $\psi \in U^*$, extend a basis of U to a basis of V, define ℓ to agree with ψ on U and be zero on the complementary basis vectors. Then $\rho(\ell) = \psi$.

By rank-nullity:

$$\dim(V^*) = \dim(\mathrm{Ann}(U)) + \dim(U^*).$$

Since $\dim(V^*) = \dim(V)$ and $\dim(U^*) = \dim(U)$, we get $\dim(\mathrm{Ann}(U)) = \dim(V) - \dim(U)$. □

Theorem 7.0.2 Annihilator isomorphism

For a subspace $U \subseteq V$, there is a natural isomorphism

$$\mathrm{Ann}(U) \cong (V/U)^*.$$

Proof: Define $\Phi : \mathrm{Ann}(U) \to (V/U)^*$ by $\Phi(\ell)(v + U) = \ell(v)$.

Well-defined: If $v + U = v' + U$, then $v - v' \in U$, so $\ell(v) - \ell(v') = \ell(v - v') = 0$ since $\ell \in \mathrm{Ann}(U)$.

Injective: If $\Phi(\ell) = 0$, then $\ell(v) = 0$ for all $v \in V$, so $\ell = 0$.

Surjective: Given $\psi \in (V/U)^*$, define $\ell = \psi \circ q$ where $q : V \to V/U$ is the quotient map. Then $\ell \in \mathrm{Ann}(U)$ and $\Phi(\ell) = \psi$. □

7.1 Quotient Spaces

Quotient spaces provide a systematic way to "collapse" a subspace to zero, creating a smaller vector space that captures the structure "perpendicular" to the subspace. This construction is fundamental in linear algebra and pervades abstract algebra.

7.1.1 Construction of the Quotient Space

Definition 7.1.1: Cosets and quotient space

Let V be a vector space over k and $U \subseteq V$ a subspace. For $v \in V$, the **coset** of v modulo U is

$$v + U = \{v + u : u \in U\}.$$

The **quotient space** is the set of all cosets:

$$V/U = \{v + U : v \in V\}.$$

Note:-

Two cosets $v + U$ and $w + U$ are equal if and only if $v - w \in U$. Equivalently, the cosets partition V into equivalence classes under the relation $v \sim w \Leftrightarrow v - w \in U$.

Proposition 7.1.1 Vector space structure on the quotient

The quotient V/U becomes a vector space over k under the operations:

1. **Addition:** $(v + U) + (w + U) = (v + w) + U$.
2. **Scalar multiplication:** $\lambda(v + U) = (\lambda v) + U$.

The zero element is $0 + U = U$, and $-(v + U) = (-v) + U$.

Proof: The key is well-definedness: if $v + U = v' + U$ and $w + U = w' + U$, then $v - v' \in U$ and $w - w' \in U$, so $(v + w) - (v' + w') = (v - v') + (w - w') \in U$, meaning $(v + w) + U = (v' + w') + U$.

Similarly for scalar multiplication: if $v + U = v' + U$, then $v - v' \in U$, so $\lambda(v - v') = \lambda v - \lambda v' \in U$, giving $\lambda v + U = \lambda v' + U$.

The vector space axioms follow from those in V. $\square$

7.1.2 The Quotient Map and Exact Sequences

Definition 7.1.2: Quotient map

The **quotient map** (or **canonical projection**) is $q : V \to V/U$ defined by $q(v) = v + U$.

Proposition 7.1.2 Properties of the quotient map

The quotient map $q : V \to V/U$ satisfies:

1. q is linear.
2. q is surjective.
3. $\ker(q) = U$.

Proof: (1) $q(v + \lambda w) = (v + \lambda w) + U = (v + U) + \lambda(w + U) = q(v) + \lambda q(w)$.

(2) Every coset $v + U$ is $q(v)$.

(3) $v \in \ker(q) \Leftrightarrow v + U = 0 + U \Leftrightarrow v \in U$. $\square$

Note:-

This gives a **short exact sequence**:

$$0 \longrightarrow U \xrightarrow{\iota} V \xrightarrow{q} V/U \longrightarrow 0$$

where $\iota : U \hookrightarrow V$ is the inclusion. "Exact" means: $\mathrm{Im}(\iota) = \ker(q)$, ι is injective, and q is surjective. Short exact sequences are a central organizing principle in algebra.

Theorem 7.1.1 Dimension of the quotient space

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(V/U) = \dim(V) - \dim(U).$$

Proof: Apply rank-nullity to $q : V \to V/U$:

$$\dim(V) = \dim(\ker(q)) + \dim(\mathrm{Im}(q)) = \dim(U) + \dim(V/U).$$

$\square$

Example 7.1.1 (Concrete example)

Let $V = \mathbb{R}^3$ and $U = \mathrm{Span}\{(1,0,0)\}$, the x-axis. Then $V/U \cong \mathbb{R}^2$:
The coset $(a,b,c) + U$ consists of all points $\{(a+t, b, c) : t \in \mathbb{R}\}$—a horizontal line through (a,b,c). Two points have the same coset iff they have the same y and z coordinates. The map $(a,b,c) + U \mapsto (b,c)$ is an isomorphism $V/U \xrightarrow{\sim} \mathbb{R}^2$.

7.1.3 Universal Property of the Quotient

The quotient space is characterized by a universal property that explains its importance.

Theorem 7.1.2 Universal property of the quotient

Let $\varphi : V \to W$ be a linear map with $U \subseteq \ker(\varphi)$. Then there exists a unique linear map $\bar{\varphi} : V/U \to W$ such that $\varphi = \bar{\varphi} \circ q$:

$$\begin{tikzcd} V \arrow[r, "\varphi"] \arrow[d, "q"'] & W \\ V/U \arrow[ur, dashed, "\exists! \bar{\varphi}"'] & \end{tikzcd}$$

Moreover, $\bar{\varphi}$ is defined by $\bar{\varphi}(v + U) = \varphi(v)$.

Proof: **Well-defined:** If $v + U = v' + U$, then $v - v' \in U \subseteq \ker(\varphi)$, so $\varphi(v) = \varphi(v')$.

Linearity: $\bar{\varphi}((v+U)+\lambda(w+U)) = \bar{\varphi}((v+\lambda w)+U) = \varphi(v+\lambda w) = \varphi(v)+\lambda\varphi(w) = \bar{\varphi}(v+U)+\lambda\bar{\varphi}(w+U)$.

Uniqueness: Any $\psi : V/U \to W$ with $\varphi = \psi \circ q$ must satisfy $\psi(v+U) = \psi(q(v)) = \varphi(v) = \bar{\varphi}(v+U)$. $\square$

Corollary 7.1.1 First Isomorphism Theorem

For any linear map $\varphi : V \to W$:

$$V/\ker(\varphi) \cong \mathrm{Im}(\varphi).$$

The isomorphism is $\bar{\varphi} : v + \ker(\varphi) \mapsto \varphi(v)$.

Proof: By the universal property, $\bar{\varphi} : V/\ker(\varphi) \to W$ is well-defined. Its image is exactly $\mathrm{Im}(\varphi)$. Injectivity: if $\bar{\varphi}(v + \ker(\varphi)) = 0$, then $\varphi(v) = 0$, so $v \in \ker(\varphi)$, meaning $v + \ker(\varphi) = 0 + \ker(\varphi)$. □

Note:-

This theorem is remarkably useful. It says that every surjective linear map $\varphi : V \twoheadrightarrow W$ is "really" a quotient map: $V \to V/\ker(\varphi) \cong W$. Dually, every injective map $\iota : U \hookrightarrow V$ makes U isomorphic to a subspace of V.

7.1.4 Correspondence Between Subspaces

The quotient construction establishes a bijection between certain subspaces.

Theorem 7.1.3 Lattice correspondence

Let $U \subseteq V$ be a subspace. There is a bijection

$$\{\text{subspaces } W \subseteq V \text{ with } U \subseteq W\} \longleftrightarrow \{\text{subspaces of } V/U\}$$

given by:

- Forward: $W \mapsto W/U = \{w + U : w \in W\}$.
- Backward: $\bar{W} \mapsto q^{-1}(\bar{W}) = \{v \in V : v + U \in \bar{W}\}$.

Moreover, this bijection preserves inclusions: $W_1 \subseteq W_2 \Leftrightarrow W_1/U \subseteq W_2/U$.

Proof: We verify the maps are inverses.

Forward then backward: Starting with $W \supseteq U$, we get W/U, then $q^{-1}(W/U) = \{v \in V : v + U \in W/U\}$. But $v + U \in W/U$ iff $v + U = w + U$ for some $w \in W$, iff $v - w \in U \subseteq W$, iff $v \in W$. So $q^{-1}(W/U) = W$.

Backward then forward: Starting with $\bar{W} \subseteq V/U$, let $W = q^{-1}(\bar{W})$. Note $U = q^{-1}(0) \subseteq q^{-1}(\bar{W}) = W$. Then $W/U = q(W) = q(q^{-1}(\bar{W})) = \bar{W}$ (surjectivity of q ensures $q(q^{-1}(S)) = S$ for any $S \subseteq V/U$). □

Note:-

Maximal proper subspaces of V containing U correspond to maximal proper subspaces of V/U—i.e., hyperplanes (codimension-1 subspaces). This perspective becomes useful in algebraic geometry and representation theory.

7.2 Linear Operators and Endomorphisms

When the domain and codomain of a linear map coincide, we gain additional algebraic structure: the ability to compose maps with themselves.

7.2.1 The Ring of Endomorphisms

Definition 7.2.1: Linear operator / Endomorphism

A **linear operator** or **endomorphism** of a vector space V is a linear map $\varphi : V \to V$. We write

$$\mathrm{End}(V) = \mathrm{Hom}(V, V)$$

for the set of all endomorphisms of V.

Proposition 7.2.1 Ring structure on $\mathrm{End}(V)$

$\mathrm{End}(V)$ is a **ring** (with identity) under:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$.
- **Multiplication:** $(\varphi \circ \psi)(v) = \varphi(\psi(v))$ (composition).
- **Identity:** The identity map id_V.

Moreover, $\mathrm{End}(V)$ is a k-algebra: multiplication distributes over scalar multiplication.

Proof: Addition makes $\mathrm{End}(V)$ an abelian group (it's a subgroup of $\mathrm{Hom}(V, V)$ as a vector space). Composition is associative with identity id_V. Distributivity: $\varphi \circ (\psi + \rho) = \varphi \circ \psi + \varphi \circ \rho$ and $(\varphi + \psi) \circ \rho = \varphi \circ \rho + \psi \circ \rho$. These follow from linearity. □

Note:-

The ring $\mathrm{End}(V)$ is noncommutative unless $\dim(V) \leqslant 1$. For $V = k^n$, $\mathrm{End}(V) \cong M_n(k)$, the ring of $n \times n$ matrices. The noncommutativity of matrix multiplication reflects the noncommutativity of $\mathrm{End}(V)$.

7.2.2 Polynomials of Operators

Since endomorphisms can be composed, we can form "powers" and evaluate polynomials.

Definition 7.2.2: Powers and polynomials of operators

For $\varphi \in \mathrm{End}(V)$ and $n \in \mathbb{Z}_{\geqslant 0}$:

$$\varphi^0 = \mathrm{id}_V, \quad \varphi^n = \underbrace{\varphi \circ \varphi \circ \cdots \circ \varphi}_{n \text{ times}}.$$

For a polynomial $p(x) = a_0 + a_1 x + \cdots + a_n x^n \in k[x]$, define

$$p(\varphi) = a_0 \mathrm{id}_V + a_1 \varphi + \cdots + a_n \varphi^n \in \mathrm{End}(V).$$

Proposition 7.2.2 Evaluation homomorphism

The map $\mathrm{ev}_\varphi : k[x] \to \mathrm{End}(V)$ defined by $p(x) \mapsto p(\varphi)$ is a ring homomorphism. That is:

1. $(p + q)(\varphi) = p(\varphi) + q(\varphi)$.
2. $(pq)(\varphi) = p(\varphi) \circ q(\varphi)$.
3. $1(\varphi) = \mathrm{id}_V$.

Proof: (1) and (3) are immediate. For (2), let $p(x) = \sum a_i x^i$ and $q(x) = \sum b_j x^j$. Then $(pq)(x) = \sum_k c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$. So

$$(pq)(\varphi) = \sum_k c_k \varphi^k = \sum_k \left(\sum_{i+j=k} a_i b_j \right) \varphi^k = \left(\sum_i a_i \varphi^i \right) \circ \left(\sum_j b_j \varphi^j \right) = p(\varphi) \circ q(\varphi).$$

□

Note:-

This makes $\mathrm{End}(V)$ a $k[x]$-module: $p(x) \cdot v = p(\varphi)(v)$. The study of this module structure (particularly the kernel of ev_φ, called the minimal polynomial) is central to the theory of canonical forms.

7.3 Invariant Subspaces

Understanding an operator often begins with finding subspaces on which its behavior is simpler.

Definition 7.3.1: Invariant subspace

A subspace $W \subseteq V$ is **invariant** under $\varphi \in \mathrm{End}(V)$ (or φ**-invariant**) if

$$\varphi(W) \subseteq W,$$

i.e., $\varphi(w) \in W$ for all $w \in W$.

Example 7.3.1 (Basic examples of invariant subspaces)

For any $\varphi \in \mathrm{End}(V)$:

- $\{0\}$ and V are always φ-invariant.
- $\ker(\varphi)$ is φ-invariant: if $\varphi(v) = 0$, then $\varphi(\varphi(v)) = \varphi(0) = 0$.
- $\mathrm{Im}(\varphi)$ is φ-invariant: $\varphi(\mathrm{Im}(\varphi)) = \mathrm{Im}(\varphi^2) \subseteq \mathrm{Im}(\varphi)$.
- More generally, $\ker(\varphi^n)$ and $\mathrm{Im}(\varphi^n)$ are φ-invariant for all $n \geqslant 0$.

Proposition 7.3.1 Restriction to an invariant subspace

If $W \subseteq V$ is φ-invariant, the **restriction** $\varphi|_W : W \to W$ is a well-defined linear operator on W.

7.3.1 Block Diagonal Structure

Invariant subspaces lead to simpler matrix representations.

Theorem 7.3.1 Direct sum of invariant subspaces

Suppose $V = V_1 \oplus V_2$ where both V_1 and V_2 are φ-invariant. Choose bases $\mathcal{B}_1 = \{e_1, \dots, e_m\}$ of V_1 and $\mathcal{B}_2 = \{e_{m+1}, \dots, e_n\}$ of V_2, forming a basis $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ of V. Then the matrix of φ with respect to $\mathcal{B}$ is **block diagonal**:

$$\mathcal{M}(\varphi; \mathcal{B}) = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

where $A_1 = \mathcal{M}(\varphi|_{V_1}; \mathcal{B}_1)$ and $A_2 = \mathcal{M}(\varphi|_{V_2}; \mathcal{B}_2)$.

Proof: For $i \leqslant m$: $\varphi(e_i) \in V_1$ (by φ-invariance), so $\varphi(e_i)$ is a linear combination of $e_1, \dots, e_m$ only. For $i > m$: $\varphi(e_i) \in V_2$, so $\varphi(e_i)$ is a linear combination of $e_{m+1}, \dots, e_n$ only. This gives the block structure. □

Note:-

Conversely, if a matrix is block diagonal with blocks A_1 and A_2, then the spans of the corresponding basis vectors are invariant subspaces. The goal of much of spectral theory is to decompose V into a direct sum of invariant subspaces on which φ acts simply.

Corollary 7.3.1 Block triangular from one invariant subspace

If V_1 is φ-invariant (but we don't assume V_1 has an invariant complement), then extending a basis of V_1 to a basis of V gives a **block upper-triangular** matrix:

$$\mathcal{M}(\varphi) = \begin{pmatrix} A_1 & * \\ 0 & A_2 \end{pmatrix}.$$

The upper-right block $*$ may be nonzero; it measures the "interaction" between V_1 and V/V_1.

7.4 Eigenvalues and Eigenvectors

The simplest invariant subspaces are one-dimensional. These give rise to the fundamental notion of eigenvalues.

7.4.1 Definitions and Basic Properties

Definition 7.4.1: Eigenvector and eigenvalue

Let $\varphi \in \mathrm{End}(V)$. A nonzero vector $v \in V$ is an **eigenvector** of φ if

$$\varphi(v) = \lambda v$$

for some scalar $\lambda \in k$. The scalar λ is called the **eigenvalue** corresponding to v.

Note:-

The condition $\varphi(v) = \lambda v$ says that φ acts on the line spanned by v as scaling by λ. Equivalently, $\mathrm{Span}\{v\}$ is a one-dimensional φ-invariant subspace. Conversely, any one-dimensional invariant subspace $U = \mathrm{Span}\{v\}$ consists of eigenvectors (with the same eigenvalue) plus zero.

Proposition 7.4.1 Characterization of eigenvalues

For $\varphi \in \mathrm{End}(V)$ and $\lambda \in k$, the following are equivalent:

1. λ is an eigenvalue of φ.
2. $\varphi - \lambda \mathrm{id}_V$ is not injective.
3. $\ker(\varphi - \lambda \mathrm{id}_V) \neq \{0\}$.
4. (If V is finite-dimensional) $\varphi - \lambda \mathrm{id}_V$ is not surjective.
5. (If V is finite-dimensional) $\det(\varphi - \lambda \mathrm{id}_V) = 0$.

Proof: (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3): λ is an eigenvalue iff there exists nonzero v with $\varphi(v) = \lambda v$, iff $(\varphi - \lambda \mathrm{id})(v) = 0$ for some nonzero v, iff $\ker(\varphi - \lambda \mathrm{id}) \neq \{0\}$.

(3) $\Leftrightarrow$ (4) $\Leftrightarrow$ (5): In finite dimensions, a linear operator is injective iff surjective iff invertible iff has nonzero determinant. So $\ker(\varphi - \lambda \mathrm{id}) \neq \{0\}$ iff $\det(\varphi - \lambda \mathrm{id}) = 0$. ☐

Definition 7.4.2: Eigenspace

For an eigenvalue λ of φ, the **eigenspace** is

$$E(\lambda, \varphi) = \ker(\varphi - \lambda \mathrm{id}_V) = \{v \in V : \varphi(v) = \lambda v\}.$$

This is a subspace of V (it includes 0, though 0 is not an eigenvector).

Theorem 7.4.1 Eigenvectors with distinct eigenvalues are linearly independent

Let $v_1, \ldots, v_m$ be eigenvectors of φ with distinct eigenvalues $\lambda_1, \ldots, \lambda_m$. Then $v_1, \ldots, v_m$ are linearly independent.

Proof: Induction on m. Base case $m = 1$: a single eigenvector is nonzero, hence linearly independent. Suppose the result holds for $m-1$ eigenvectors. Assume $\sum_{i=1}^{m} c_i v_i = 0$. Apply φ:

$$\sum_{i=1}^{m} c_i \lambda_i v_i = 0.$$

Multiply the original equation by λ_m and subtract:

$$\sum_{i=1}^{m-1} c_i(\lambda_i - \lambda_m) v_i = 0.$$

By the induction hypothesis (since $v_1, \dots, v_{m-1}$ are eigenvectors with distinct eigenvalues $\lambda_1, \dots, \lambda_{m-1}$), we have $c_i(\lambda_i - \lambda_m) = 0$ for $i < m$. Since $\lambda_i \neq \lambda_m$, we get $c_i = 0$. Substituting back: $c_m v_m = 0$, so $c_m = 0$ (since $v_m \neq 0$). □

Corollary 7.4.1 Bound on number of eigenvalues

A linear operator on an n-dimensional space has at most n distinct eigenvalues.

Proof: If $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues with eigenvectors $v_1, \dots, v_m$, then $v_1, \dots, v_m$ are linearly independent. In an n-dimensional space, any linearly independent set has at most n elements. □

Corollary 7.4.2 Sum of eigenspaces is direct

If $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of φ, then

$$E(\lambda_1, \varphi) + \cdots + E(\lambda_m, \varphi) = E(\lambda_1, \varphi) \oplus \cdots \oplus E(\lambda_m, \varphi).$$

is a direct sum. Furthermore, $\sum_i \dim E(\lambda_i, \varphi) \leqslant \dim V$.

7.4.2 Diagonalizable Operators

The best-case scenario: V has a basis of eigenvectors.

Definition 7.4.3: Diagonalizable operator

An operator $\varphi \in \mathrm{End}(V)$ is **diagonalizable** if there exists a basis of V consisting entirely of eigenvectors of φ. Equivalently, φ has a diagonal matrix with respect to some basis.

Note:-

If $\{v_1, \dots, v_n\}$ is a basis of eigenvectors with $\varphi(v_i) = \lambda_i v_i$, then the matrix of φ is

$$\mathcal{M}(\varphi) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

The eigenvalues appear on the diagonal (possibly with repetition).

Theorem 7.4.2 Characterizations of diagonalizability

For $\varphi \in \mathrm{End}(V)$ with $\dim(V) = n$ and distinct eigenvalues $\lambda_1, \dots, \lambda_m$, the following are equivalent:

1. φ is diagonalizable.
2. V has a basis of eigenvectors of φ.
3. $V = E(\lambda_1, \varphi) \oplus \cdots \oplus E(\lambda_m, \varphi)$.

4. $\dim \mathsf{E}(\lambda_1, \varphi) + \cdots + \dim \mathsf{E}(\lambda_m, \varphi) = n$.

Proof: (1) $\Leftrightarrow$ (2): By definition.

(2) $\Rightarrow$ (3): If V has a basis of eigenvectors, partition it by eigenvalue. Each part is a basis of the corresponding eigenspace, and together they span V.

(3) $\Rightarrow$ (4): Take dimensions: $\dim V = \dim(\mathsf{E}_1 \oplus \cdots \oplus \mathsf{E}_m) = \sum \dim \mathsf{E}_i$.

(4) $\Rightarrow$ (2): The sum $\mathsf{E}_1 + \cdots + \mathsf{E}_m$ is direct (earlier corollary), and has dimension $\sum \dim \mathsf{E}_i = n = \dim V$. So $\mathsf{E}_1 \oplus \cdots \oplus \mathsf{E}_m = V$. Combining bases of each E_i gives a basis of V consisting of eigenvectors. □

Corollary 7.4.3 Enough distinct eigenvalues implies diagonalizability

If $\varphi \in \mathrm{End}(V)$ has $\dim(V)$ distinct eigenvalues, then φ is diagonalizable.

Proof: Each eigenspace has dimension at least 1 (it contains an eigenvector). With $n = \dim V$ distinct eigenvalues, $\sum \dim \mathsf{E}(\lambda_i) \geqslant n$. But this sum is also $\leqslant n$, so equality holds, and criterion (4) above is satisfied. □

7.4.3 Examples: Diagonalizable and Non-Diagonalizable

Example 7.4.1 (Diagonal matrices)

The identity matrix I_n has eigenvalue 1 with eigenspace all of k^n. More generally, a diagonal matrix $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ has the standard basis vectors as eigenvectors with eigenvalues $\lambda_1, \ldots, \lambda_n$. Such a matrix is (trivially) diagonalizable—it's already diagonal!

Example 7.4.2 (A non-diagonalizable matrix)

Consider $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues satisfy $\det(A - \lambda I) = (1 - \lambda)^2 = 0$, so $\lambda = 1$ is the only eigenvalue.

The eigenspace: $\mathsf{E}(1, \varphi) = \ker(A - I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \mathrm{Span}\{e_1\}$.

This has dimension 1, but $\dim(\mathbb{R}^2) = 2$. So φ is **not diagonalizable**.
Geometrically, φ is a "shear": it fixes the x-axis pointwise and slides other points parallel to the x-axis. The only eigenvector direction is along the x-axis.

Example 7.4.3 (No eigenvectors over $\mathbb{R}$)

Consider rotation by 90° in $\mathbb{R}^2$:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic polynomial is $\det(R - \lambda I) = \lambda^2 + 1$, which has no real roots. So R has <u>no eigenvalues</u> over $\mathbb{R}$, hence no eigenvectors. Over $\mathbb{C}$, the eigenvalues are $\pm i$ with eigenvectors $(1, \mp i)^T$, and R is diagonalizable as a complex matrix. The eigenfairy doesn't visit $\mathbb{R}$ (sad).
This illustrates that diagonalizability depends on the scalar field!

Note:-

The rotation example shows that not every operator is diagonalizable. Even over $\mathbb{C}$, where every polynomial has roots, an operator may fail to be diagonalizable (like the shear above). Understanding non-diagonalizable operators requires the theory of Jordan normal form, which we'll develop later.

Question 1: Exercises on quotient spaces, operators, and eigenvalues

1. Let $V = \mathbb{R}^4$ and $U = \{(x, y, z, w) : x + y = 0, z = w\}$. Find $\dim(U)$, $\dim(V/U)$, and describe V/U explicitly.
2. Prove that if $\varphi, \psi \in \mathrm{End}(V)$ commute ($\varphi \circ \psi = \psi \circ \varphi$), then every eigenspace of φ is ψ-invariant.
3. Let $\varphi \in \mathrm{End}(V)$ be diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_m$. Prove that $p(\varphi) = 0$ where $p(x) = \prod_{i=1}^{m}(x - \lambda_i)$.
4. Show that the quotient map $q : V \to V/U$ induces a bijection between {linear maps $V/U \to W$} and {linear maps $V \to W$ vanishing on U}.
5. Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Find the eigenvalues and eigenspaces of A. Is A diagonalizable? If so, find a matrix P such that $P^{-1}AP$ is diagonal.

Solution

(1) The constraints $x + y = 0$ and $z = w$ give 2 independent linear equations on $\mathbb{R}^4$, so $\dim(U) = 4 - 2 = 2$. A basis: $\{(-1, 1, 0, 0), (0, 0, 1, 1)\}$.
By the dimension formula: $\dim(V/U) = \dim(V) - \dim(U) = 4 - 2 = 2$.
The map $\pi : \mathbb{R}^4 \to \mathbb{R}^2$ defined by $\pi(x, y, z, w) = (x + y, z - w)$ has $\ker(\pi) = U$. By the first isomorphism theorem, $V/U \cong \mathbb{R}^2$. Explicitly, the coset $(x, y, z, w) + U$ is determined by $(x + y, z - w)$.
(2) Let v be an eigenvector of φ with eigenvalue λ: $\varphi(v) = \lambda v$. We need $\psi(v) \in E(\lambda, \varphi)$, i.e., $\varphi(\psi(v)) = \lambda\psi(v)$.
Compute: $\varphi(\psi(v)) = \psi(\varphi(v)) = \psi(\lambda v) = \lambda\psi(v)$. ✓
(We used commutativity in the first equality.)
(3) Let $p(x) = \prod_{i=1}^{m}(x - \lambda_i)$. Since φ is diagonalizable, $V = \bigoplus_{i=1}^{m} E(\lambda_i, \varphi)$. For any eigenvector $v \in E(\lambda_j, \varphi)$:

$$(\varphi - \lambda_j \mathrm{id})(v) = 0.$$

So $p(\varphi)(v) = (\varphi - \lambda_1) \cdots (\varphi - \lambda_m)(v) = 0$ because one factor gives zero.
Since this holds for all eigenvectors and they span V, we have $p(\varphi) = 0$.
(4) Define $\Phi : \mathrm{Hom}(V/U, W) \to \{T \in \mathrm{Hom}(V, W) : U \subseteq \ker(T)\}$ by $\Phi(\bar{T}) = \bar{T} \circ q$.
<u>Injective:</u> If $\bar{T} \circ q = 0$, then $\bar{T}(v + U) = \bar{T}(q(v)) = 0$ for all v, so $\bar{T} = 0$.
<u>Surjective:</u> Given $T : V \to W$ with $U \subseteq \ker(T)$, define $\bar{T} : V/U \to W$ by $\bar{T}(v + U) = T(v)$. Well-defined because $v - v' \in U \Rightarrow T(v) = T(v')$. Then $\Phi(\bar{T}) = \bar{T} \circ q = T$.
(5) Characteristic polynomial: $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$.
Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.
Eigenspaces:

- $\lambda = 5$: $(A - 5I)v = 0$ gives $\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = 2y$. Eigenspace: $\mathrm{Span}\{(2, 1)^T\}$.
- $\lambda = 2$: $(A - 2I)v = 0$ gives $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = -y$. Eigenspace: $\mathrm{Span}\{(1, -1)^T\}$.

Since we have 2 linearly independent eigenvectors in $\mathbb{R}^2$, A is diagonalizable.
Let $P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ (columns are eigenvectors). Then $P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$.

7.5 Let's get general up in here!

7.5.1 Generalized Kernals (Null Spaces)

We want to look at general eigenvectors: things that are not *actually* eigenvectors but "kinda want to be" in a generalized sense. What in tarnation does that mean!? Simplest Eigenspace is just 0, so let's focus on that to get an understanding:

Definition 7.5.1: Generalized kernal

The generalized kernal of φ is $\text{gker}(\varphi) = \{v \in V \mid \exists m > 0 \text{ s.t. } \varphi^m(v) = 0\}$. These are all the vecrtors that are evnetually sent to 0 by repeatedly applying φ.

Note:-

We note that $\text{gker}(\varphi)$ is not standard notation. Auroux likes it though, so that is what we are going with! *If you meet a stranger on the street and bring up gker be sure to tell them what you mean, lest they be confused.*

The difference between the kernal and generalized kernal can be summed up by this: you are in the kernal if you go to 0 in one step; if you apply itterative steps—φ^m—and end up at 0 you are in the generalized kernal. If I look at the kernal of φ I notice that it is in the kernal of φ^2; then too this holds for more! They may or may not get larger but they are subsets:

$$\text{Ker}(\varphi) \subset \text{Ker}(\varphi^2) \subset \text{Ker}\,\varphi \subset \dots$$

Proposition 7.5.1

If $\text{Ker}(\varphi^m) = \text{Ker}(\varphi^{m+1})$ then there is no strictly larger kernal.

Proof: $\varphi^{m+2}(v) = \varphi^{m+1}(\varphi(V)) = 0 \iff \varphi(v) = \text{Ker}\,\varphi^m \iff \varphi^m(\varphi(v)) = \varphi^{m+1}(v) = 0$. So $\text{Ker}(\varphi^{m+1}) = \text{Ker}(\varphi^{m+2})$. Since the sequence stops increasing after at most $n = \dim V$ steps, $\text{gker}(\varphi) = \text{Ker}\,\varphi^n$. $\square$

Example 7.5.1

$\varphi : k^2 \to k^2$ with $e_1 \mapsto 0$, $e_2 \mapsto e_1$ and $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, then $\text{Ker}(\varphi) = k \cdot e_1$, but $\text{Ker}(\varphi^2) = \text{gker}(\varphi) = k^2$.

Note:-

The generalized kernal is often referred to as the null space (this is the terminology that Axler uses, for example). I may use this interchangeably.

Since it is not the case that $V = \text{gker}(\varphi) \oplus \text{Img}(\varphi)$ for every $\varphi \in \text{Hom}(V, V)$, we can use the following proposition as a substitute:

Proposition 7.5.2 Generalized kernel-image decomposition

Suppose $\varphi \in \text{End}(V)$ and let m be the smallest positive integer such that $\text{Ker}(\varphi^m) = \text{Ker}(\varphi^{m+1})$. Then

$$V = \text{Ker}(\varphi^m) \oplus \text{Im}(\varphi^m).$$

In particular, $V = \text{gker}(\varphi) \oplus \text{Im}(\varphi^{\dim V})$.

Proof: We verify the two conditions for a direct sum decomposition.

Intersection is trivial: Suppose $v \in \text{Ker}(\varphi^m) \cap \text{Im}(\varphi^m)$. Write $v = \varphi^m(u)$ for some $u \in V$. Then $\varphi^m(v) = \varphi^{2m}(u) = 0$, so $u \in \text{Ker}(\varphi^{2m})$.

We claim $\operatorname{Ker}(\varphi^{2m}) = \operatorname{Ker}(\varphi^m)$. By the stabilization property, once the kernel sequence stabilizes at step m, it stays constant: $\operatorname{Ker}(\varphi^m) = \operatorname{Ker}(\varphi^{m+1}) = \cdots = \operatorname{Ker}(\varphi^{2m})$. Thus $u \in \operatorname{Ker}(\varphi^m)$, which means $v = \varphi^m(u) = 0$.

Sum is all of V: By rank-nullity applied to φ^m:

$$\dim V = \dim \operatorname{Ker}(\varphi^m) + \dim \operatorname{Im}(\varphi^m).$$

Since we just showed $\operatorname{Ker}(\varphi^m) \cap \operatorname{Im}(\varphi^m) = \{0\}$, the sum $\operatorname{Ker}(\varphi^m) + \operatorname{Im}(\varphi^m)$ is direct, and its dimension equals $\dim \operatorname{Ker}(\varphi^m) + \dim \operatorname{Im}(\varphi^m) = \dim V$. Hence it equals V. □

Note:-

This decomposition is the key structural result for understanding non-diagonalizable operators. The generalized kernel $\operatorname{gker}(\varphi)$ captures the "nilpotent part" of φ, while $\operatorname{Im}(\varphi^m)$ is where φ acts invertibly. On $\operatorname{gker}(\varphi)$, repeated application of φ eventually kills everything; on $\operatorname{Im}(\varphi^m)$, the restriction of φ is an isomorphism.

Corollary 7.5.1 Restriction to image is an isomorphism

With notation as above, the restriction $\varphi|_{\operatorname{Im}(\varphi^m)} : \operatorname{Im}(\varphi^m) \to \operatorname{Im}(\varphi^m)$ is an isomorphism.

Proof: First, $\operatorname{Im}(\varphi^m)$ is φ-invariant since $\varphi(\operatorname{Im}(\varphi^m)) = \operatorname{Im}(\varphi^{m+1}) \subseteq \operatorname{Im}(\varphi^m)$. For injectivity: if $v \in \operatorname{Im}(\varphi^m)$ and $\varphi(v) = 0$, then $v \in \operatorname{Ker}(\varphi) \subseteq \operatorname{Ker}(\varphi^m)$. But $\operatorname{Ker}(\varphi^m) \cap \operatorname{Im}(\varphi^m) = \{0\}$, so $v = 0$. Since $\varphi|_{\operatorname{Im}(\varphi^m)}$ is injective on a finite-dimensional space, it's also surjective onto its image, which equals $\operatorname{Im}(\varphi^m)$ by φ-invariance. □

7.5.2 Generalized Eigenspaces

Now we extend the idea of generalized kernels to arbitrary eigenvalues, not just $\lambda = 0$.

Definition 7.5.2: Generalized eigenspace

Let $\varphi \in \operatorname{End}(V)$ and let λ be an eigenvalue of φ. The **generalized eigenspace** of φ corresponding to λ is

$$V_\lambda = \operatorname{Ker}(\varphi - \lambda \mathrm{id})^{\dim V} = \{v \in V : (\varphi - \lambda \mathrm{id})^n(v) = 0 \text{ for some } n \geqslant 1\}.$$

Equivalently, $V_\lambda = \operatorname{gker}(\varphi - \lambda \mathrm{id})$.

Note:-

The ordinary eigenspace $E(\lambda, \varphi) = \operatorname{Ker}(\varphi - \lambda \mathrm{id})$ is always contained in V_λ, but V_λ may be strictly larger. The elements of $V_\lambda \setminus E(\lambda, \varphi)$ are called **generalized eigenvectors**—they are not true eigenvectors, but they are "eventually killed" by $\varphi - \lambda \mathrm{id}$.

Example 7.5.2 (Generalized eigenspace of the shear)

Recall the shear $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The only eigenvalue is $\lambda = 1$, and the eigenspace is $E(1, \varphi) = \operatorname{Ker}(A - I) = \operatorname{Ker} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \operatorname{Span}\{e_1\}$, which is one-dimensional.

However, $(A - I)^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$.

So the generalized eigenspace is $V_1 = \operatorname{Ker}(A - I)^2 = \mathbb{R}^2$. The vector e_2 is a generalized eigenvector: $(A - I)e_2 = e_1 \neq 0$, but $(A - I)^2 e_2 = 0$.

Proposition 7.5.3 Basic properties of generalized eigenspaces

Let $\varphi \in \operatorname{End}(V)$ with eigenvalue λ. Then:

1. $E(\lambda, \varphi) \subseteq V_\lambda$.
2. V_λ is φ-invariant.
3. On V_λ, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent.
4. If $\mu \neq \lambda$ are distinct eigenvalues, then $V_\lambda \cap V_\mu = \{0\}$.

Proof: (1) is immediate: if $(\varphi - \lambda\mathrm{id})(v) = 0$, then certainly $(\varphi - \lambda\mathrm{id})^n(v) = 0$ for all $n \geqslant 1$.

(2) Let $v \in V_\lambda$, so $(\varphi - \lambda\mathrm{id})^n(v) = 0$ for some n. Since φ commutes with $\varphi - \lambda\mathrm{id}$, we have

$$(\varphi - \lambda\mathrm{id})^n(\varphi(v)) = \varphi((\varphi - \lambda\mathrm{id})^n(v)) = \varphi(0) = 0.$$

Thus $\varphi(v) \in V_\lambda$.

(3) By definition, $V_\lambda = \mathrm{Ker}(\varphi - \lambda\mathrm{id})^m$ for some $m \leqslant \dim V$. So $(\varphi - \lambda\mathrm{id})^m$ restricted to V_λ is the zero map.

(4) Suppose $v \in V_\lambda \cap V_\mu$ with $\lambda \neq \mu$. Then there exist $j, k \geqslant 1$ such that $(\varphi - \lambda\mathrm{id})^j(v) = 0$ and $(\varphi - \mu\mathrm{id})^k(v) = 0$.

The polynomials $p(x) = (x - \lambda)^j$ and $q(x) = (x - \mu)^k$ are coprime (they share no common roots). By Bézout's identity, there exist polynomials $a(x), b(x) \in k[x]$ such that $a(x)p(x) + b(x)q(x) = 1$.

Evaluating at φ: $a(\varphi)p(\varphi) + b(\varphi)q(\varphi) = \mathrm{id}$. Applying to v:

$$v = a(\varphi)p(\varphi)(v) + b(\varphi)q(\varphi)(v) = a(\varphi)(0) + b(\varphi)(0) = 0.$$

□

Note:-

Part (4) shows that generalized eigenspaces for distinct eigenvalues intersect trivially, just like ordinary eigenspaces. The proof uses the coprimality of $(x-\lambda)^j$ and $(x-\mu)^k$—this is a technique that will reappear in the generalized eigenspace decomposition.

7.5.3 The Generalized Eigenspace Decomposition

The following theorem is the main structural result for operators on finite-dimensional vector spaces over algebraically closed fields.

Theorem 7.5.1 Generalized eigenspace decomposition

Let V be a finite-dimensional vector space over an algebraically closed field k, and let $\varphi \in \mathrm{End}(V)$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of φ. Then

$$V = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_m}.$$

Moreover, $\sum_{i=1}^m \dim V_{\lambda_i} = \dim V$.

Proof: We proceed in two steps.

Step 1: The sum is direct. We already showed that $V_{\lambda_i} \cap V_{\lambda_j} = \{0\}$ for $i \neq j$. More generally, suppose $v_1 + \cdots + v_m = 0$ where $v_i \in V_{\lambda_i}$. We show each $v_i = 0$.

For each i, let n_i be such that $(\varphi - \lambda_i\mathrm{id})^{n_i}(v_i) = 0$. Define $p_i(x) = (x - \lambda_i)^{n_i}$ and

$$q_i(x) = \prod_{j \neq i}(x - \lambda_j)^{n_j}.$$

Since λ_i is not a root of $q_i(x)$, the polynomials $p_i(x)$ and $q_i(x)$ are coprime. By Bézout, there exist $a_i(x), b_i(x)$ with $a_i(x)p_i(x) + b_i(x)q_i(x) = 1$.

Now, $q_i(\varphi)(v_j) = 0$ for all $j \neq i$ (since p_j divides q_i). Applying $q_i(\varphi)$ to $v_1 + \cdots + v_m = 0$:

$$0 = q_i(\varphi)(v_1 + \cdots + v_m) = q_i(\varphi)(v_i).$$

Thus $v_i \in \mathrm{Ker}(q_i(\varphi)) \cap \mathrm{Ker}(p_i(\varphi)) = V_{\lambda_i} \cap \mathrm{Ker}(q_i(\varphi))$.

But from $a_i(\varphi)p_i(\varphi)(v_i) + b_i(\varphi)q_i(\varphi)(v_i) = v_i$ and both terms being zero, we get $v_i = 0$.

Step 2: The sum equals V. Define $p(x) = \prod_{i=1}^m (x - \lambda_i)^{n_i}$ where $n_i = \dim V_{\lambda_i}$ (we'll justify this choice shortly). We claim $p(\varphi) = 0$, i.e., p annihilates φ.

Consider the polynomial $q_i(x) = p(x)/(x-\lambda_i)^{n_i}$. These polynomials are pairwise coprime (each q_i excludes exactly the factor $(x - \lambda_i)^{n_i}$). Their gcd is 1, so there exist $r_i(x)$ with $\sum_i r_i(x)q_i(x) = 1$.

For any $v \in V$, write $v = \sum_i r_i(\varphi)q_i(\varphi)(v)$. We claim $r_i(\varphi)q_i(\varphi)(v) \in V_{\lambda_i}$. Indeed:

$$p_i(\varphi)(r_i(\varphi)q_i(\varphi)(v)) = r_i(\varphi)(p_i(\varphi)q_i(\varphi))(v) = r_i(\varphi)p(\varphi)(v).$$

Once we establish that $p(\varphi) = 0$ (which follows from Cayley-Hamilton, proved later, or can be verified by dimension counting), this shows v is a sum of elements from the V_{λ_i}.

Alternative dimension argument: Since k is algebraically closed, φ has at least one eigenvalue. Apply the decomposition $V = \mathrm{gker}(\varphi - \lambda_1 \mathrm{id}) \oplus \mathrm{Im}((\varphi - \lambda_1 \mathrm{id})^{n_1})$ where n_1 is the stabilization index. The first summand is V_{λ_1}. On the second summand, $\varphi - \lambda_1 \mathrm{id}$ is invertible, so λ_1 is not an eigenvalue of φ restricted there. Proceed by induction on $\dim V$. □

Corollary 7.5.2 Dimension of generalized eigenspaces

With notation as above, if $\chi_\varphi(x) = \prod_{i=1}^m (x-\lambda_i)^{n_i}$ is the characteristic polynomial, then $\dim V_{\lambda_i} = n_i$.

Note:-

The exponent n_i in the characteristic polynomial is called the **algebraic multiplicity** of λ_i. The dimension of the ordinary eigenspace $\dim E(\lambda_i, \varphi)$ is called the **geometric multiplicity**. We always have:

$$1 \leqslant \dim E(\lambda_i, \varphi) \leqslant \dim V_{\lambda_i} = n_i.$$

The operator φ is diagonalizable if and only if geometric multiplicity equals algebraic multiplicity for every eigenvalue.

Example 7.5.3 (Decomposition for a 3×3 matrix)

Consider $\varphi : \mathbb{C}^3 \to \mathbb{C}^3$ with matrix

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

The characteristic polynomial is $\chi_A(x) = (x-2)^2(x-3)$, so $\lambda_1 = 2$ has algebraic multiplicity 2 and $\lambda_2 = 3$ has algebraic multiplicity 1.
For $\lambda_1 = 2$:

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (A-2I)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

So $V_2 = \mathrm{Ker}(A-2I)^2 = \mathrm{Span}\{e_1, e_2\}$, which has dimension 2.
For $\lambda_2 = 3$: $V_3 = \mathrm{Ker}(A - 3I) = \mathrm{Span}\{e_3\}$, dimension 1.
Indeed, $\mathbb{C}^3 = V_2 \oplus V_3$.

7.6 Nilpotent Operators

Nilpotent operators are the building blocks for understanding non-diagonalizable maps. On each generalized eigenspace V_λ, the operator $\varphi - \lambda \mathrm{id}$ is nilpotent, so understanding nilpotent operators is the key to understanding all operators.

7.6.1 Definition and Basic Properties

Definition 7.6.1: Nilpotent operator

An operator $\varphi \in \mathrm{End}(V)$ is **nilpotent** if $\varphi^m = 0$ for some positive integer m. The smallest such m is called the **index of nilpotency** (or **nilpotence index**) of φ.

Example 7.6.1 (Basic examples of nilpotent operators)

1. The zero operator is nilpotent with index 1.
2. The shift operator on k^n with matrix $\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$ is nilpotent with index n.
3. Any strictly upper-triangular matrix (zeros on and below the diagonal) is nilpotent.

Proposition 7.6.1 Characterizations of nilpotent operators

For $\varphi \in \mathrm{End}(V)$ with $\dim V = n$, the following are equivalent:

1. φ is nilpotent.
2. $\varphi^n = 0$.
3. The only eigenvalue of φ is 0 (if k is algebraically closed).
4. $V = \mathrm{gker}(\varphi)$.

Proof: (1) $\Rightarrow$ (2): If $\varphi^m = 0$, then the chain $\{0\} \subseteq \mathrm{Ker}(\varphi) \subseteq \mathrm{Ker}(\varphi^2) \subseteq \cdots$ stabilizes by step m, hence by step n (since dimensions are bounded by n). So $\mathrm{Ker}(\varphi^n) = \mathrm{gker}(\varphi)$. But $\varphi^m = 0$ means $\mathrm{Ker}(\varphi^m) = V$, so $\mathrm{gker}(\varphi) = V$, which means $\varphi^n = 0$.

(2) $\Rightarrow$ (3): Suppose λ is an eigenvalue with eigenvector $v \neq 0$. Then $\varphi(v) = \lambda v$, so $\varphi^n(v) = \lambda^n v$. If $\varphi^n = 0$, then $\lambda^n v = 0$, so $\lambda^n = 0$, hence $\lambda = 0$.

(3) $\Rightarrow$ (4): Over an algebraically closed field, by the generalized eigenspace decomposition, $V = \bigoplus_\lambda V_\lambda$. If the only eigenvalue is 0, then $V = V_0 = \mathrm{gker}(\varphi)$.

(4) $\Rightarrow$ (1): If $V = \mathrm{gker}(\varphi) = \mathrm{Ker}(\varphi^n)$, then $\varphi^n = 0$. $\square$

Note:-

Over a non-algebraically closed field, (3) needs modification: φ is nilpotent iff φ has no eigenvalues in any field extension, or equivalently, iff the characteristic polynomial is x^n.

Corollary 7.6.1 Nilpotent matrices are similar to strictly upper-triangular matrices

If φ is nilpotent, then there exists a basis in which the matrix of φ is strictly upper-triangular.

Proof: We prove this by induction on $n = \dim V$. If $n = 1$, then $\varphi = 0$ and we're done.

For $n > 1$: since φ is nilpotent and nonzero (otherwise trivial), $\mathrm{Ker}(\varphi) \neq \{0\}$. Pick $0 \neq v_1 \in \mathrm{Ker}(\varphi)$ and extend to a basis $\{v_1, v_2, \ldots, v_n\}$. The matrix of φ has first column all zeros.

Alternatively, consider $W = \mathrm{Im}(\varphi)$. Since φ is nilpotent and nonzero, $W \subsetneq V$ (otherwise φ would be surjective, hence bijective, contradicting nilpotency). The restriction $\varphi|_W : W \to W$ is well-defined (W is

φ-invariant) and nilpotent. By induction, choose a basis of W making $\varphi|_W$ strictly upper-triangular, extend to a basis of V, and the result follows. $\square$

7.6.2 The Nilpotent Basis Construction

The key to understanding nilpotent operators is building a basis adapted to the "chains" of vectors connected by φ.

Definition 7.6.2: Chains and Jordan chains

Let $\varphi \in \mathrm{End}(V)$ be nilpotent. A **chain** (or **Jordan chain**) of length m is a sequence of vectors $v_1, v_2, \dots, v_m$ such that:

$$\varphi(v_m) = v_{m-1}, \quad \varphi(v_{m-1}) = v_{m-2}, \quad \dots, \quad \varphi(v_2) = v_1, \quad \varphi(v_1) = 0.$$

The vector v_m is called the **head** of the chain, and v_1 is the **tail**.

Note:-

In a chain, $v_1 \in \mathrm{Ker}(\varphi)$, $v_2 \in \mathrm{Ker}(\varphi^2) \setminus \mathrm{Ker}(\varphi)$, and generally $v_j \in \mathrm{Ker}(\varphi^j) \setminus \mathrm{Ker}(\varphi^{j-1})$. The chain encodes how φ "moves" vectors toward the kernel.

Lemma 7.6.1 Key lemma for chain construction

Let φ be nilpotent. Suppose $U \subseteq \mathrm{Ker}(\varphi^{k+1})$ is a subspace satisfying $U \cap \mathrm{Ker}(\varphi^k) = \{0\}$. Then:

1. $\varphi|_U$ is injective.
2. $\varphi(U) \subseteq \mathrm{Ker}(\varphi^k)$.
3. $\varphi(U) \cap \mathrm{Ker}(\varphi^{k-1}) = \{0\}$ (if $k \geqslant 1$).

Proof: (1) If $v \in U$ and $\varphi(v) = 0$, then $v \in U \cap \mathrm{Ker}(\varphi) \subseteq U \cap \mathrm{Ker}(\varphi^k) = \{0\}$.

(2) For $v \in U \subseteq \mathrm{Ker}(\varphi^{k+1})$, we have $\varphi^{k+1}(v) = 0$, so $\varphi^k(\varphi(v)) = 0$, meaning $\varphi(v) \in \mathrm{Ker}(\varphi^k)$.

(3) If $\varphi(v) \in \mathrm{Ker}(\varphi^{k-1})$, then $\varphi^k(v) = \varphi^{k-1}(\varphi(v)) = 0$, so $v \in \mathrm{Ker}(\varphi^k)$. By assumption $U \cap \mathrm{Ker}(\varphi^k) = \{0\}$, so $v = 0$, hence $\varphi(v) = 0$. $\square$

Theorem 7.6.1 Structure theorem for nilpotent operators

Let $\varphi \in \mathrm{End}(V)$ be nilpotent with index m (so $\varphi^m = 0$ but $\varphi^{m-1} \neq 0$). Then there exists a basis of V of the form

$$\{v_{1,1}, \dots, v_{1,m_1};\ v_{2,1}, \dots, v_{2,m_2};\ \dots;\ v_{\ell,1}, \dots, v_{\ell,m_\ell}\}$$

where each sequence $(v_{i,1}, \dots, v_{i,m_i})$ is a chain:

$$\varphi(v_{i,j}) = v_{i,j-1} \text{ for } j > 1, \quad \varphi(v_{i,1}) = 0.$$

The chain lengths satisfy $m = m_1 \geqslant m_2 \geqslant \cdots \geqslant m_\ell \geqslant 1$ and $\sum_{i=1}^{\ell} m_i = \dim V$.

Proof: We construct the basis by working down from $\mathrm{Ker}(\varphi^m) = V$.

Step 1: Let U_m be a complement of $\mathrm{Ker}(\varphi^{m-1})$ in $V = \mathrm{Ker}(\varphi^m)$. Pick a basis $\{v_{1,m}, \dots, v_{k_m,m}\}$ of U_m. These will be the heads of chains of length m.

By the lemma, $\varphi(U_m) \subseteq \mathrm{Ker}(\varphi^{m-1})$ and $\varphi(U_m) \cap \mathrm{Ker}(\varphi^{m-2}) = \{0\}$. Define $v_{i,m-1} = \varphi(v_{i,m})$ for each i. These vectors are linearly independent (since $\varphi|_{U_m}$ is injective).

Step 2: Now work in $\mathrm{Ker}(\varphi^{m-1})$. We have $\varphi(U_m) \subseteq \mathrm{Ker}(\varphi^{m-1})$ with $\varphi(U_m) \cap \mathrm{Ker}(\varphi^{m-2}) = \{0\}$. Extend $\varphi(U_m)$ to a larger subspace U_{m-1} that complements $\mathrm{Ker}(\varphi^{m-2})$ in $\mathrm{Ker}(\varphi^{m-1})$. Pick additional basis vectors $\{v_{k_m+1,m-1}, \dots, v_{k_{m-1},m-1}\}$ to span the complement of $\varphi(U_m)$ in U_{m-1}. These start new chains of length $m-1$.

Continue inductively: At each stage, we have chains of various lengths. Applying φ extends existing chains, and we add new chain heads as needed to complement $\mathrm{Ker}(\varphi^{j-1})$ in $\mathrm{Ker}(\varphi^j)$.

Termination: At the final step, we're in $\mathrm{Ker}(\varphi)$, and all chains terminate. The collection of all chain vectors forms a basis of V. □

Note:-

The chain lengths $(m_1, m_2, \dots, m_\ell)$ form a partition of $n = \dim V$. This partition is an invariant of the nilpotent operator—two nilpotent operators are similar iff they have the same partition.

7.6.3 Jordan Blocks for Nilpotent Operators

The chain basis gives the matrix of a nilpotent operator a very specific form.

Definition 7.6.3: Nilpotent Jordan block

A **nilpotent Jordan block** of size m is the $m \times m$ matrix

$$J_m(0) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

This has 1's on the superdiagonal and 0's elsewhere.

Corollary 7.6.2 Matrix form of nilpotent operators

With respect to a chain basis ordered as $(v_{1,1}, \dots, v_{1,m_1}, v_{2,1}, \dots, v_{2,m_2}, \dots)$, the matrix of a nilpotent operator φ is block diagonal:

$$\mathcal{M}(\varphi) = \begin{pmatrix} J_{m_1}(0) & & 0 \\ & J_{m_2}(0) & \\ 0 & & \ddots \end{pmatrix}.$$

Proof: Within each chain $(v_{i,1}, \dots, v_{i,m_i})$, we have $\varphi(v_{i,j}) = v_{i,j-1}$ for $j > 1$ and $\varphi(v_{i,1}) = 0$. In matrix form with this basis ordering, this gives exactly $J_{m_i}(0)$ for the i-th block. The blocks are independent since chains span complementary subspaces. □

Example 7.6.2 (Nilpotent operator with partition $(3, 2, 1)$)

Consider $\varphi : k^6 \to k^6$ nilpotent with chain lengths $3, 2, 1$. A chain basis might be:

$$\{v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{3,1}\}$$

where $\varphi(v_{1,3}) = v_{1,2}$, $\varphi(v_{1,2}) = v_{1,1}$, $\varphi(v_{1,1}) = 0$, etc.
The matrix is:

$$\mathcal{M}(\varphi) = \begin{pmatrix} 0 & 1 & 0 & & & \\ 0 & 0 & 1 & & 0 & \\ 0 & 0 & 0 & & & \\ & & & 0 & 1 & \\ & 0 & & 0 & 0 & \\ & & & & & 0 \end{pmatrix} = J_3(0) \oplus J_2(0) \oplus J_1(0).$$

Proposition 7.6.2 Uniqueness of the Jordan form for nilpotent operators

The partition $(m_1, m_2, \ldots, m_\ell)$ associated to a nilpotent operator φ is uniquely determined by φ. Equivalently, two nilpotent operators are similar iff they have the same Jordan block sizes (counting multiplicity).

Proof: The partition can be recovered from dimensions of iterated kernels. Let $d_j = \dim \operatorname{Ker}(\varphi^j)$. Then:

$$d_1 - d_0 = d_1 = \text{number of chains} = \ell$$

$$d_2 - d_1 = \text{number of chains of length} \geqslant 2$$

$$d_j - d_{j-1} = \text{number of chains of length} \geqslant j.$$

From this sequence of differences, we can reconstruct the partition: the number of blocks of size exactly j is $(d_j - d_{j-1}) - (d_{j+1} - d_j)$. Since the d_j are determined by φ, so is the partition. ☐

Note:-

This dimension-counting argument is the key to proving uniqueness of Jordan form in general. The rank of $(\varphi - \lambda I)^j$ determines how many Jordan blocks for λ have size $\geqslant j$.

7.7 Jordan Normal Form

We now combine the generalized eigenspace decomposition with the structure theory of nilpotent operators to obtain the Jordan normal form—the "best possible" matrix representation for an operator over an algebraically closed field.

7.7.1 Jordan Blocks

Definition 7.7.1: Jordan block

A **Jordan block** of size m with eigenvalue λ is the $m \times m$ matrix

$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix} = \lambda I_m + J_m(0).$$

This has λ on the diagonal, 1's on the superdiagonal, and 0's elsewhere.

Example 7.7.1 (Small Jordan blocks)

$$J_1(\lambda) = (\lambda), \quad J_2(\lambda) = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \quad J_3(\lambda) = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}.$$

A 1×1 Jordan block is just a scalar—this corresponds to a true eigenvector. Larger blocks correspond to generalized eigenvectors that are not true eigenvectors.

Proposition 7.7.1 Properties of Jordan blocks

Let $J = J_m(\lambda)$. Then:

1. The only eigenvalue of J is λ, with algebraic multiplicity m.

2. The eigenspace $E(\lambda, J) = \mathrm{Ker}(J - \lambda I) = \mathrm{Span}\{e_1\}$ has dimension 1.
3. The generalized eigenspace is all of k^m.
4. $J - \lambda I = J_m(0)$ is nilpotent with index m.

Proof: (1) The characteristic polynomial is $\det(J - xI) = (\lambda - x)^m$.

(2) $J - \lambda I = J_m(0)$ has kernel spanned by e_1 (the first standard basis vector).

(3) Since $(J - \lambda I)^m = J_m(0)^m = 0$, every vector is in $\mathrm{Ker}(J - \lambda I)^m$.

(4) $J_m(0)^{m-1} \neq 0$ (it maps $e_m \mapsto e_1$) but $J_m(0)^m = 0$. □

Note:-

A Jordan block $J_m(\lambda)$ represents the action of an operator on a single chain of generalized eigenvectors. The chain is $(e_1, e_2, \ldots, e_m)$ where $Je_j = \lambda e_j + e_{j-1}$ for $j > 1$ and $Je_1 = \lambda e_1$. Equivalently, $(J - \lambda I)e_j = e_{j-1}$ for $j > 1$ and $(J - \lambda I)e_1 = 0$.

7.7.2 The Jordan Normal Form Theorem

Theorem 7.7.1 Jordan normal form

Let V be a finite-dimensional vector space over an algebraically closed field k, and let $\varphi \in \mathrm{End}(V)$. Then there exists a basis of V with respect to which the matrix of φ is block diagonal:

$$\mathcal{M}(\varphi) = \begin{pmatrix} J_{m_1}(\lambda_1) & & 0 \\ & J_{m_2}(\lambda_2) & \\ 0 & & \ddots \end{pmatrix}$$

where each block is a Jordan block. This matrix is called the **Jordan normal form** (or **Jordan canonical form**) of φ.

Proof: By the generalized eigenspace decomposition, $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_r}$ where $\lambda_1, \ldots, \lambda_r$ are the distinct eigenvalues. Each V_{λ_i} is φ-invariant, so we can work on each generalized eigenspace separately.

On V_{λ_i}, the operator $\psi_i = \varphi|_{V_{\lambda_i}} - \lambda_i \mathrm{id}$ is nilpotent (this was proved earlier). By the structure theorem for nilpotent operators, there exists a chain basis of V_{λ_i} such that ψ_i has matrix $J_{m_1}(0) \oplus J_{m_2}(0) \oplus \cdots$.

Since $\varphi|_{V_{\lambda_i}} = \psi_i + \lambda_i \mathrm{id}$, the matrix of $\varphi|_{V_{\lambda_i}}$ in this basis is

$$J_{m_1}(0) + \lambda_i I \oplus J_{m_2}(0) + \lambda_i I \oplus \cdots = J_{m_1}(\lambda_i) \oplus J_{m_2}(\lambda_i) \oplus \cdots$$

Combining the bases for all generalized eigenspaces gives a basis of V in which φ has the desired block diagonal form. □

Note:-

The Jordan form groups blocks by eigenvalue: all $J_m(\lambda_i)$ blocks for a given λ_i appear together, corresponding to the decomposition of V_{λ_i} into chains. The total size of blocks with eigenvalue λ_i equals $\dim V_{\lambda_i}$ = algebraic multiplicity of λ_i.

7.7.3 Existence and Uniqueness

Theorem 7.7.2 Uniqueness of Jordan normal form

The Jordan normal form of φ is unique up to permutation of the blocks. That is, the multiset of Jordan blocks $\{J_{m_1}(\lambda_1), J_{m_2}(\lambda_2), \ldots\}$ is uniquely determined by φ.

Proof: For each eigenvalue λ, the Jordan blocks with eigenvalue λ correspond to the partition of $\dim V_\lambda$ given by the nilpotent operator $\varphi|_{V_\lambda} - \lambda\mathrm{id}$. We showed this partition is uniquely determined by dimensions of iterated kernels:

$$d_j(\lambda) = \dim \operatorname{Ker}((\varphi - \lambda\mathrm{id})^j).$$

The number of blocks $J_m(\lambda)$ of size exactly m is

$$(d_m(\lambda) - d_{m-1}(\lambda)) - (d_{m+1}(\lambda) - d_m(\lambda)) = 2d_m(\lambda) - d_{m-1}(\lambda) - d_{m+1}(\lambda).$$

Since these dimensions are intrinsic to φ, the block structure is uniquely determined. □

Corollary 7.7.1 Similarity criterion

Two operators $\varphi, \psi \in \operatorname{End}(V)$ are similar if and only if they have the same Jordan normal form (up to block ordering). Equivalently, $\varphi \sim \psi$ iff for each $\lambda \in k$ and each $j \geqslant 1$:

$$\dim \operatorname{Ker}((\varphi - \lambda\mathrm{id})^j) = \dim \operatorname{Ker}((\psi - \lambda\mathrm{id})^j).$$

Note:-

This gives a complete set of similarity invariants. Unlike trace and determinant alone (which don't distinguish all similarity classes), the full sequence of kernel dimensions completely determines the similarity class.

7.7.4 Computing Jordan Form: The Algorithm

Given a matrix A, here is the procedure to find its Jordan form:

Step 1: Find eigenvalues. Compute $\chi_A(x) = \det(xI - A)$ and factor it as $\prod_i (x - \lambda_i)^{n_i}$. The λ_i are eigenvalues with algebraic multiplicities n_i.

Step 2: For each eigenvalue λ, determine block sizes. Compute $d_j = \dim \operatorname{Ker}(A - \lambda I)^j$ for $j = 1, 2, \ldots$ until stabilization. The number of blocks of size $\geqslant j$ is $d_j - d_{j-1}$. Alternatively:

- Number of blocks $= d_1 =$ geometric multiplicity.
- Number of blocks of size $\geqslant 2$ is $d_2 - d_1$.
- Number of blocks of size exactly j is $(d_j - d_{j-1}) - (d_{j+1} - d_j)$.

Step 3: Assemble the Jordan form. List all Jordan blocks grouped by eigenvalue.

Example 7.7.2 (Computing Jordan form)

Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

Step 1: $\chi_A(x) = (x-2)^3$, so $\lambda = 2$ with multiplicity 3.

Step 2:

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A - 2I)^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A - 2I)^3 = 0.$$

So $d_1 = 1$, $d_2 = 2$, $d_3 = 3$.
Number of blocks $= d_1 = 1$. Size of the single block $= 3$.
Step 3: Jordan form is $J_3(2)$. (The matrix is already in Jordan form!)

Example 7.7.3 (A more interesting example)

Let $A = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.
Step 1: $\chi_A(x) = (x-3)^4$, so $\lambda = 3$ with multiplicity 4.
Step 2:

$$A - 3I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$\mathrm{Ker}(A - 3I) = \mathrm{Span}\{e_1, e_3\}$, so $d_1 = 2$. $(A - 3I)^2 = 0$, so $d_2 = 4$.
Number of blocks $= 2$. Blocks of size $\geqslant 2$: $d_2 - d_1 = 2$. So both blocks have size 2.
Step 3: Jordan form is $J_2(3) \oplus J_2(3)$.

Example 7.7.4 (Multiple eigenvalues)

Let $A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$.
Step 1: $\chi_A(x) = (x-1)^2(x-2)^2$. Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$, each with multiplicity 2.
Step 2 for $\lambda = 1$:

$$A - I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Restricted to generalized eigenspace (first two coordinates): $d_1 = 1$, $d_2 = 2$. One block of size 2: $J_2(1)$.
Step 2 for $\lambda = 2$:

$$A - 2I = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Restricted to generalized eigenspace (last two coordinates): $d_1 = 2$. Two blocks of size 1: $J_1(2) \oplus J_1(2)$.

Step 3: Jordan form is $J_2(1) \oplus J_1(2) \oplus J_1(2) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$.

Note:-

The last example shows that $\lambda = 2$ contributes two 1×1 blocks because the eigenspace has dimension 2 (geometric $=$ algebraic multiplicity). This is the diagonalizable case for that eigenvalue. In contrast, $\lambda = 1$ has a 2×2 block because geometric multiplicity (1) is less than algebraic multiplicity (2).

7.8 Characteristic and Minimal Polynomials

We now introduce two polynomials canonically associated to a linear operator. The characteristic polynomial encodes eigenvalues and their algebraic multiplicities; the minimal polynomial captures the "smallest" polynomial relation satisfied by the operator.

7.8.1 The Characteristic Polynomial

Definition 7.8.1: Characteristic polynomial

Let $\varphi \in \mathrm{End}(V)$ with $\dim V = n$, and let $A = \mathcal{M}(\varphi)$ be the matrix of φ with respect to some basis. The **characteristic polynomial** of φ is

$$\chi_\varphi(x) = \det(xI - \varphi) = \det(xI - A).$$

This is a monic polynomial of degree n.

Proposition 7.8.1 Basic properties of the characteristic polynomial

1. $\chi_\varphi(x)$ is independent of the choice of basis (hence well-defined for the operator φ).
2. $\chi_\varphi(x) = x^n - (\mathrm{Tr}\,\varphi)x^{n-1} + \cdots + (-1)^n \det(\varphi)$.
3. The roots of $\chi_\varphi(x)$ are exactly the eigenvalues of φ.
4. If k is algebraically closed, $\chi_\varphi(x) = \prod_{i=1}^m (x - \lambda_i)^{n_i}$ where λ_i are the distinct eigenvalues and $n_i = \dim V_{\lambda_i}$.

Proof: (1) If $B = P^{-1}AP$ is the matrix in another basis, then

$$\det(xI - B) = \det(xI - P^{-1}AP) = \det(P^{-1}(xI - A)P) = \det(xI - A).$$

(2) Expand $\det(xI - A)$ using the Leibniz formula. The leading term comes from the diagonal product $\prod_i (x - a_{ii}) = x^n - (\sum_i a_{ii})x^{n-1} + \cdots$. The constant term is $\det(-A) = (-1)^n \det(A)$.

(3) λ is a root iff $\det(\lambda I - A) = 0$ iff $\lambda I - A$ is singular iff $\mathrm{Ker}(\varphi - \lambda \mathrm{id}) \neq \{0\}$ iff λ is an eigenvalue.

(4) This follows from the generalized eigenspace decomposition: the algebraic multiplicity of λ_i equals $\dim V_{\lambda_i}$. □

Example 7.8.1 (Characteristic polynomial of a 2×2 matrix)

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\chi_A(x) = \det \begin{pmatrix} x - a & -b \\ -c & x - d \end{pmatrix} = (x-a)(x-d) - bc = x^2 - (a+d)x + (ad - bc) = x^2 - (\mathrm{Tr}\,A)x + \det A.$$

7.8.2 The Cayley-Hamilton Theorem

The Cayley-Hamilton theorem states that every operator satisfies its own characteristic polynomial.

Theorem 7.8.1 Cayley-Hamilton

Let $\varphi \in \mathrm{End}(V)$ with characteristic polynomial $\chi_\varphi(x)$. Then

$$\chi_\varphi(\varphi) = 0.$$

That is, substituting φ for x in $\chi_\varphi(x)$ gives the zero operator.

Proof: We give two proofs.

Proof 1 (via Jordan form, over algebraically closed k): It suffices to prove this for Jordan blocks, since the Jordan form is block diagonal and $\chi_\varphi = \prod \chi_{J_i}$.

For a single Jordan block $J = J_m(\lambda)$, we have $\chi_J(x) = (x-\lambda)^m$. Then

$$\chi_J(J) = (J - \lambda I)^m = J_m(0)^m = 0$$

since $J_m(0)$ is nilpotent with index m.

Proof 2 (general, using adjugate matrix): Let A be the matrix of φ. The adjugate (classical adjoint) matrix $\mathrm{adj}(xI - A)$ is a matrix whose entries are polynomials in x of degree at most $n-1$. Write

$$\mathrm{adj}(xI - A) = B_{n-1}x^{n-1} + B_{n-2}x^{n-2} + \cdots + B_1 x + B_0$$

where B_i are constant matrices.

By the adjugate identity: $(xI - A) \cdot \mathrm{adj}(xI - A) = \det(xI - A) \cdot I = \chi_A(x) \cdot I$.

Expanding both sides and comparing coefficients of powers of x gives a system of matrix equations. Multiplying the equation for x^k by A^k and summing yields $\chi_A(A) = 0$. (The details involve telescoping—each B_i appears twice with opposite signs.) $\square$

Note:-

The Cayley-Hamilton theorem does <u>not</u> follow from "plugging in A for x in $\det(xI - A) = 0$." The determinant is a scalar-valued function, not a matrix-valued function, so this substitution is meaningless. The theorem requires proof.

Corollary 7.8.1 Bound on nilpotence index

If φ is nilpotent on an n-dimensional space, then $\varphi^n = 0$.

Proof: The only eigenvalue is 0, so $\chi_\varphi(x) = x^n$. By Cayley-Hamilton, $\varphi^n = 0$. $\square$

Corollary 7.8.2 Expression for the inverse

If φ is invertible, then φ^{-1} can be expressed as a polynomial in φ.

Proof: Write $\chi_\varphi(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$. Since φ is invertible, 0 is not an eigenvalue, so $c_0 = (-1)^n \det(\varphi) \neq 0$.

By Cayley-Hamilton: $\varphi^n + c_{n-1}\varphi^{n-1} + \cdots + c_1\varphi + c_0 I = 0$.

Rearranging: $\varphi(\varphi^{n-1} + c_{n-1}\varphi^{n-2} + \cdots + c_1 I) = -c_0 I$.

Thus $\varphi^{-1} = -\frac{1}{c_0}(\varphi^{n-1} + c_{n-1}\varphi^{n-2} + \cdots + c_1 I)$. $\square$

7.8.3 The Minimal Polynomial

Definition 7.8.2: Minimal polynomial

The **minimal polynomial** $\mu_\varphi(x)$ of $\varphi \in \mathrm{End}(V)$ is the unique monic polynomial of smallest degree such that $\mu_\varphi(\varphi) = 0$.

Proposition 7.8.2 Existence and uniqueness of the minimal polynomial

1. There exists a nonzero polynomial $p(x)$ with $p(\varphi) = 0$.
2. Among all such polynomials, there is a unique monic one of minimal degree.
3. $\mu_\varphi(x)$ divides every polynomial $p(x)$ satisfying $p(\varphi) = 0$.

Proof: (1) Cayley-Hamilton gives $\chi_\varphi(\varphi) = 0$.

(2) Let d be the minimal degree of a nonzero polynomial annihilating φ. If $p(x)$ and $q(x)$ are both monic of degree d with $p(\varphi) = q(\varphi) = 0$, then $(p - q)(\varphi) = 0$ and $\deg(p - q) < d$. By minimality, $p - q = 0$.

(3) Given $p(x)$ with $p(\varphi) = 0$, perform polynomial division: $p(x) = q(x)\mu_\varphi(x) + r(x)$ where $\deg r < \deg \mu_\varphi$. Then $r(\varphi) = p(\varphi) - q(\varphi)\mu_\varphi(\varphi) = 0 - 0 = 0$. By minimality of μ_φ, we must have $r = 0$, so $\mu_\varphi \mid p$. □

Theorem 7.8.2 Relationship between characteristic and minimal polynomials

Let $\varphi \in \mathrm{End}(V)$.

1. $\mu_\varphi(x)$ divides $\chi_\varphi(x)$.
2. $\mu_\varphi(x)$ and $\chi_\varphi(x)$ have the same roots (i.e., the same irreducible factors over k).
3. Over an algebraically closed field, if $\chi_\varphi(x) = \prod_i (x - \lambda_i)^{n_i}$, then $\mu_\varphi(x) = \prod_i (x - \lambda_i)^{m_i}$ where $1 \leqslant m_i \leqslant n_i$.

Proof: (1) By Cayley-Hamilton, $\chi_\varphi(\varphi) = 0$. By the divisibility property of μ_φ, we have $\mu_\varphi \mid \chi_\varphi$.

(2) If λ is a root of χ_φ, then λ is an eigenvalue: there exists $v \neq 0$ with $\varphi(v) = \lambda v$. For any polynomial p, $p(\varphi)(v) = p(\lambda)v$. If $p(\varphi) = 0$, then $p(\lambda) = 0$. In particular, $\mu_\varphi(\lambda) = 0$.

Conversely, if $\mu_\varphi(\lambda) = 0$, then $(x - \lambda) \mid \mu_\varphi(x)$, so $(x - \lambda) \mid \chi_\varphi(x)$ by (1), meaning λ is a root of χ_φ.

(3) From (1) and (2), the factorizations have the same roots with $m_i \leqslant n_i$. The exponent m_i is the size of the largest Jordan block with eigenvalue λ_i (proof below). □

Proposition 7.8.3 Minimal polynomial and Jordan block sizes

The exponent of $(x - \lambda)$ in $\mu_\varphi(x)$ equals the size of the largest Jordan block with eigenvalue λ.

Proof: On the generalized eigenspace V_λ, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent. Its nilpotence index is the size of the largest Jordan block (the longest chain). Call this m_λ.

For $(\varphi - \lambda\mathrm{id})^k$ to annihilate V_λ, we need $k \geqslant m_\lambda$. The smallest such k is m_λ.

Since $V = \bigoplus_\lambda V_\lambda$ and these are φ-invariant, $p(\varphi) = 0$ iff $p(\varphi)|_{V_\lambda} = 0$ for all λ. The latter requires $(x - \lambda)^{m_\lambda} \mid p(x)$. Thus $\mu_\varphi(x) = \prod_\lambda (x - \lambda)^{m_\lambda}$. □

7.8.4 Diagonalizability Criterion

Theorem 7.8.3 Diagonalizability via minimal polynomial

An operator $\varphi \in \mathrm{End}(V)$ is diagonalizable if and only if $\mu_\varphi(x)$ splits into distinct linear factors:

$$\mu_\varphi(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_m)$$

where $\lambda_1, \ldots, \lambda_m$ are distinct.

Proof: ($\Rightarrow$) If φ is diagonalizable, all Jordan blocks are 1×1. The largest block for each eigenvalue has size 1, so each $(x - \lambda_i)$ appears with exponent 1 in μ_φ.

($\Leftarrow$) If $\mu_\varphi(x) = \prod_i (x-\lambda_i)$ with distinct λ_i, then each Jordan block has size 1 (since the exponent of $(x-\lambda_i)$ in μ_φ equals the largest block size for λ_i). So φ has a basis of eigenvectors. $\square$

Corollary 7.8.3 Diagonalizability of operators with distinct eigenvalues

If φ has $n = \dim V$ distinct eigenvalues, then φ is diagonalizable.

Proof: The characteristic polynomial has n distinct roots, so $\chi_\varphi(x) = \prod_{i=1}^n (x-\lambda_i)$. Since $\mu_\varphi \mid \chi_\varphi$ and they have the same roots, μ_φ also has distinct linear factors. $\square$

Example 7.8.2 (Minimal polynomial examples)

1. For $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I$: $\chi_A(x) = (x-2)^2$, but $\mu_A(x) = x-2$ since $(A-2I) = 0$.
2. For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$: $\chi_A(x) = (x-2)^2$. Since $A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$, we need $\mu_A(x) = (x-2)^2 = \chi_A(x)$.
3. For $A = \operatorname{diag}(1,1,2,2,2)$: $\chi_A(x) = (x-1)^2(x-2)^3$, but $\mu_A(x) = (x-1)(x-2)$ since $(A-I)(A-2I) = 0$.

7.8.5 Exercises

Question 2: Exercises on advanced spectral theory

1. Let $\varphi : \mathbb{C}^4 \to \mathbb{C}^4$ have characteristic polynomial $\chi_\varphi(x) = (x-1)^2(x-2)^2$. List all possible Jordan normal forms for φ.
2. Prove that if A and B are similar matrices, then $\mu_A = \mu_B$.
3. Find the Jordan normal form and minimal polynomial of $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.
4. Let $\varphi \in \operatorname{End}(V)$ satisfy $\varphi^2 = \varphi$. Prove that φ is diagonalizable and find its possible eigenvalues.
5. Suppose $\varphi \in \operatorname{End}(V)$ has minimal polynomial $\mu_\varphi(x) = (x-1)(x-2)(x-3)$. What can you conclude about φ?

Solution

(1) The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$, each with algebraic multiplicity 2. For each eigenvalue, the Jordan blocks must have sizes summing to 2. Possibilities for each:

- One block of size 2: $J_2(\lambda)$
- Two blocks of size 1: $J_1(\lambda) \oplus J_1(\lambda)$

Combining independently for each eigenvalue, the possible Jordan forms are:

(i) $J_2(1) \oplus J_2(2)$

(ii) $J_2(1) \oplus J_1(2) \oplus J_1(2)$

(iii) $J_1(1) \oplus J_1(1) \oplus J_2(2)$

(iv) $J_1(1) \oplus J_1(1) \oplus J_1(2) \oplus J_1(2)$ (diagonal matrix)

(2) If $B = P^{-1}AP$, then for any polynomial p:

$$p(B) = p(P^{-1}AP) = P^{-1}p(A)P.$$

So $p(B) = 0$ iff $p(A) = 0$. The set of annihilating polynomials is the same, hence the minimal polynomial (the monic generator of this ideal) is the same.
(3) The matrix is already in Jordan form: $A = J_3(0)$.
Characteristic polynomial: $\chi_A(x) = x^3$.

Minimal polynomial: We need the smallest k with $A^k = 0$. Compute $A^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0$ and $A^3 = 0$. So $\mu_A(x) = x^3 = \chi_A(x)$.
(4) The condition $\varphi^2 = \varphi$ means $\varphi^2 - \varphi = 0$, so $\varphi(\varphi - \mathrm{id}) = 0$. Thus $p(x) = x(x-1) = x^2 - x$ annihilates φ.
The minimal polynomial divides $x(x-1)$. Since $x(x-1)$ has distinct roots, so does μ_φ. By the diagonalizability criterion, φ is diagonalizable.
The possible eigenvalues are roots of μ_φ, which divides $x(x-1)$. So eigenvalues are among $\{0, 1\}$.
(Such operators are called **projections** or **idempotents**. They decompose $V = \mathrm{Ker}(\varphi) \oplus \mathrm{Im}(\varphi)$ and project onto $\mathrm{Im}(\varphi)$ along $\mathrm{Ker}(\varphi)$.)
(5) Since $\mu_\varphi(x) = (x-1)(x-2)(x-3)$ has three distinct linear factors:

- φ is diagonalizable (by the criterion).
- The eigenvalues are exactly $1, 2, 3$ (roots of μ_φ).
- $\dim V \geqslant 3$ (at least one eigenvector per eigenvalue).
- $V = E(1, \varphi) \oplus E(2, \varphi) \oplus E(3, \varphi)$.

The eigenspace dimensions can be any positive integers summing to $\dim V$, but we need at least one of each.